

4	(a) (i)	solid: (molecules) vibrate no translational motion / fixed position, liquid: translational motion	B1	
			B1	[2]
	(ii)	gas: molecules have random (and translational) motion	B1	[1]
	(b) (i)	ductile: straight line through origin then curving towards x-axis	B1	[1]
	(ii)	brittle: straight line through origin with no or negligible curved region	B1	[1]
	(c)	similarity: obey Hooke's law / $F \propto x$ or have elastic regions	B1	
		difference: brittle no or (very) little plastic region ductile has (large(r)) plastic region	B1	[2]